

Module 04: Cognition — The Prediction Machine

Learning Objectives

1. Understand that your **Brain** is a **Prediction Machine**.
2. Learn that "Thinking" means guessing what will happen next.
3. Know the difference between a **Good Prediction** and a **Surprise**.

Introduction: The Prediction Machine

Your Brain is like a super-fast computer. It is always asking: "What happens next?"

- If I drop this ball... **Prediction:** It will fall down!
- If I touch this hot stove... **Prediction:** Ouch!
- If I see a cloud... **Prediction:** Maybe rain?

Captain Brain uses these predictions to keep the Body Ship safe.

Key Concepts

1. The Model (Memory Map)

Your brain remembers everything that happened before. It builds a map called a **Generative Model**. "Last time I ate a lemon, it was sour." "Next time I see a lemon, I predict... SOUR!"

2. Matching (Good Prediction)

If the ball falls down, your brain is happy. "Yay! I was right! The world makes sense."

3. Surprise (Oops!)

If the ball floats UP... Whoa! **Surprise!** Your brain wakes up! "That's weird! I need to learn!"
Surprise helps us learn new things.

Activities

Activity 1: The Magic Bag

Put 3 blue blocks and 1 red block in a bag. **Predict:** If I pull one out, what color will it be? (Probably Blue!) Pull one out. If it is Red... **Surprise!** Your brain has to update its guess.

Activity 2: Pattern Power

Clap-Clap-Stomp. Clap-Clap-Stomp. Clap-Clap-...? What comes next? (Stomp!) Your brain learned the pattern and predicted the future!

Summary

Thinking is really just **Predicting**. Your Brain uses memories to guess the future. When you guess right, you are safe. When you are surprised, you learn something new!

References

- *If You Give a Mouse a Cookie* by Laura Numeroff (A story about what happens next!)
- *Thinking, Fast and Slow* (For parents/teachers)